

Lecture 7 : Continuity

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Theorem 1 Let $D \subseteq \mathbb{R}$ be a set and $f : D \rightarrow \mathbb{R}$ be continuous function at $c \in D$. Then, $|f|$ is continuous at c .

Proof: $|f| : D \rightarrow \mathbb{R}$ is defined by $|f|(x) = |f(x)|$, $x \in D$.

$$||f|(x) - |f|(c)| = ||f(x) - f(c)|| \leq |f(x) - f(c)|.$$

Let us choose, $\epsilon > 0$.

Since f is continuous at c , there exists a positive δ , such that

$$|f(x) - f(c)| < \epsilon \text{ for all } x \in N(c, \delta) \cap D.$$

Therefore,

$$||f|(x) - |f|(c)| < \epsilon \text{ for all } x \in N(c, \delta) \cap D.$$

This shows that, $|f|$ is continuous at c .

Theorem 2 Let $A, B \subseteq \mathbb{R}$ and $f : A \rightarrow \mathbb{R}$, $g : B \rightarrow \mathbb{R}$ be functions such that $f(A) \subset B$. Let $c \in A$ and f is continuous at c and g is continuous at $f(c) \in B$. Then the composite function $g \circ f : A \rightarrow \mathbb{R}$ is continuous at c .

Proof: Let W be a neighborhood of $g \circ f(c)$.

Since, g is continuous at $f(c)$, there exists a neighborhood V of $f(c)$ such that $y \in V \cap B \Rightarrow g(y) \in W$.

Since, f is continuous at c , corresponding to the neighborhood V of $f(c)$, there exists a neighborhood U of c , such that $x \in U \cap A \Rightarrow f(x) \in V$.

Since, $f(A) \subset B$, $f(x) \in V \Rightarrow f(x) \in V \cap B$.

Therefore, $x \in U \cap A \Rightarrow f(x) \in V$

$$\Rightarrow f(x) \in V \cap B$$

$$\Rightarrow g \circ f(x) \in W.$$

Thus, corresponding to a chosen neighborhood W of $g \circ f(c)$, there exists a neighborhood U of c such that for all $x \in U \cap A$, $g \circ f(x) \in W$.

This proves that, composite function $g \circ f(x)$ is continuous at c .

1. Discontinuity

A function is discontinuous at a point $x = c$, if the function is not continuous at c . We shall now list the different types of discontinuities.

1. Removable discontinuity

A function f has a removable discontinuity at a point c , if $\lim_{x \rightarrow c} f(x)$ exists but

$$\lim_{x \rightarrow c} f(x) \neq f(c).$$

This continuity can be removed by changing the value of $f(c)$, such that

$$\lim_{x \rightarrow c} f(x) = f(c)$$

Example : Let

$$f(x) = \begin{cases} \frac{x^2 - 4}{x - 2}, & x \neq 2 \\ 10, & x = 2 \end{cases}$$

Here, $\lim_{x \rightarrow 2} f(x) = 4$, but $f(2) = 10$. Thus, $\lim_{x \rightarrow 2} f(x) \neq f(2)$.

Hence, f is not continuous at 2 and 2 is a point of removable discontinuity.

2. Discontinuity of first kind

A function f is said to have a discontinuity of first kind at a point c , if limits from both the left and right exist, but not equal. i.e.,

$$\lim_{x \rightarrow c^+} f(x) \neq \lim_{x \rightarrow c^-} f(x).$$

- (a) **Discontinuity from left:** A function f is said to have a discontinuity of first kind from left at a point c , if

$$\lim_{x \rightarrow c^-} f(x) \neq f(c), \text{ but } \lim_{x \rightarrow c^+} f(x) = f(c).$$

Example : Let

$$f(x) = \begin{cases} x, & x \geq 2 \\ -1, & x < 2 \end{cases}$$

Here, $\lim_{x \rightarrow 2^+} f(x) = 2$, $\lim_{x \rightarrow 2^-} f(x) = -1$ and $f(2) = 2$. Thus, $\lim_{x \rightarrow 2^+} f(x) = f(2) = 2$ and $\lim_{x \rightarrow 2^-} f(x) \neq f(2)$.

Hence, f has discontinuity of first kind from left.

- (b) **Discontinuity from right:** A function f is said to have a discontinuity of first kind from right at a point c , if

$$\lim_{x \rightarrow c^+} f(x) \neq f(c), \text{ but } \lim_{x \rightarrow c^-} f(x) = f(c).$$

Example : Let

$$f(x) = \begin{cases} x^2, & x \leq 2 \\ -1, & x > 2 \end{cases}$$

Here, $\lim_{x \rightarrow 2^-} f(x) = 4$, $\lim_{x \rightarrow 2^+} f(x) = -1$ and $f(2) = 4$. Thus, $\lim_{x \rightarrow 2^-} f(x) = f(2) = 4$ and $\lim_{x \rightarrow 2^+} f(x) \neq f(2)$.

Hence, f has discontinuity of first kind from right.

3. Discontinuity of the second kind A function f is said to have a discontinuity of second kind at a point c , if either of the limit from left or right does not exist.

(a) **Discontinuity from left:** A function f is said to have a discontinuity of second kind from left at a point c , if $\lim_{x \rightarrow c^-} f(x)$ does not exist.

Example : Let

$$f(x) = \begin{cases} 1, & x \geq 2 \\ \sin\left(\frac{1}{x-2}\right), & x < 2 \end{cases}.$$

Here, $\lim_{x \rightarrow 2^+} f(x) = 1$, $\lim_{x \rightarrow 2^-} f(x)$ does not exist.

Hence, f has discontinuity of second kind from left.

(b) **Discontinuity from right:** A function f is said to have a discontinuity of second kind from right at a point c , if $\lim_{x \rightarrow c^+} f(x)$ does not exist.

Example : Let

$$f(x) = \begin{cases} 1, & x < 2 \\ \sin\left(\frac{1}{x-2}\right), & x \geq 2 \end{cases}.$$

Here, $\lim_{x \rightarrow 2^-} f(x) = 1$ and, $\lim_{x \rightarrow 2^+} f(x)$ does not exist.

Hence, f has discontinuity of second kind from right.

Theorem 3 Let $I = [a, b]$ be a closed and bounded interval and a function $f : I \rightarrow \mathbb{R}$ be continuous on I . Then, f is bounded on I .

Proof: Let f is an unbounded and continuous function on I .

Then, there exists some $x_r \in [a, b]$, such that $|f(x_r)| \geq M$, for very large $M > 0$.

Since, f is unbounded, then there exists some $x_k \in [a, b]$, such that $|f(x_k)| \rightarrow \infty$ when $x \rightarrow x_k$.

Therefore, $|f(x_k) - f(x_r)| \rightarrow \infty$, but $|x_k - x_r| \leq b - a$. This implies, f is not continuous, which is contradiction. Thus, if f is continuous on I , then f is bounded on I .

Theorem 4 Let $I = [a, b]$ be a closed and bounded interval and a function $f : I \rightarrow \mathbb{R}$ be continuous on I . Then, there exists a point c in I , such that $f(c) = \sup_{x \in I} f(x)$ and also, there is a point d in I , such that $f(d) = \inf_{x \in I} f(x)$.

Proof: The set $f(I) = \{f(x) : x \in I\}$ is a bounded set. Since, this is a non-empty bounded subset of $\mathbb{R}$, $\sup f(I)$ and $\inf f(I)$ exists.

Let $M = \sup f(I)$, $m = \inf f(I)$. Then, for all $x \in [a, b]$, $m \leq f(x) \leq M$.

We shall prove that $f(x) = M$ for some $c \in [a, b]$.

If not, let $f(x) < M$ for all $x \in [a, b]$. Then, $M - f(x) > 0$, for all $x \in [a, b]$.

Let $\phi(x) = \frac{1}{M - f(x)}$, $x \in [a, b]$. Then, ϕ is continuous on $[a, b]$ and therefore ϕ is bounded on $[a, b]$.

Let B be an upper bound of ϕ on $[a, b]$. Then, $B > 0$ and $0 < \frac{1}{M - f(x)} < B$ for all $x \in [a, b]$.

Therefore, $f(x) < M - \frac{1}{B}$ for all $x \in [a, b]$. This contradicts that, $M = \sup f(I)$.

Hence, there exists a point c in $[a, b]$, such that $f(c) = M$.

In a similar manner, it can be prove that there exists a point d in $[a, b]$ such that $f(d) = m$. This completes the proof.

Theorem 5 : Bolzano's Theorem

Let $[a, b]$ be a closed and bounded interval and $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$. If $f(a)$ and $f(b)$ are of opposite signs then there exists at least a point c in the open interval (a, b) such that $f(c) = 0$.

(For proof, see the reference book.)

Theorem 6 : Intermediate value theorem.

Let $[a, b]$ be a closed and bounded interval and a function $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$. If $f(a) \neq f(b)$, then f attains every value between $f(a)$ and $f(b)$ at least once in the open interval (a, b) .

Proof: Without loss of generality, we assume that $f(a) < f(b)$.

Let μ be a real number such that $f(a) < \mu < f(b)$.

Let us consider the function $\phi : [a, b] \rightarrow \mathbb{R}$ defined by $\phi(x) = f(x) - \mu$, $x \in [a, b]$.

ϕ is continuous on $[a, b]$, since f is continuous on $[a, b]$.

$\phi(a) = f(a) - \mu < 0$, $\phi(b) = f(b) - \mu > 0$.

As $\phi(a)$ and $\phi(b)$ are of opposite signs, by Bolzano's theorem there exists at least one point c in (a, b) such that $\phi(c) = 0$.

Therefore, $f(c) - \mu = 0$, i.e., $f(c) = \mu$.

Thus f attains μ at a point c in (a, b) and the theorem is done.

Example : A function $f : [0, 1] \rightarrow [0, 1]$ is continuous on $[0, 1]$. Prove that there exists

a point c in $[0, 1]$, such that $f(c) = c$.

Solution: If $f(0) = 0$ or $f(1) = 1$, the existence is proved.

We assume, $f(0) \neq 0$ and $f(1) \neq 1$.

Let us consider the function $g : [0, 1] \rightarrow \mathbb{R}$ defined by $g(x) = f(x) - x$, $x \in [0, 1]$.

g is continuous on $[0, 1]$ and $g(0) = f(0) > 0$, since $f(0) \in [0, 1]$ and $f(0) \neq 0$. Also $g(1) = f(1) - 1 < 0$, since $f(1) \in [0, 1]$ and $f(1) \neq 1$.

By the Intermediate value theorem there exists a point c in $(0, 1)$ such that $g(c) = 0$. Therefore, $f(c) = c$.

Reference book : *Introduction to real analysis by S.K. Mapa.*